Experiment 1 – Optical free-beam transmission system

Explanation: goal, principle, procedure and why we need it later

1. The experiment in one sentence

An LED sends light through the air to a photodiode: we measure the characteristics of both components and the transmission behaviour of the whole system – efficiency, distance dependence, rise time, cut-off frequency – and finally transmit music.

2. Goal of the experiment

- Record the LED I-V curve and determine the forward voltage U_D.
- Measure the photodiode current versus distance r; give P_{PD} in W and dBm.
- Determine the overall efficiency $\eta = P_{PD}/P_{LED}$ at r_{min}.
- Determine rise/fall time (20 kHz square wave), signal delay and phase (1–80 kHz) and the cut-off frequency (−3 dB).
- Build a music transmission over ≥ 1 m; compare analog and digital transmission.

3. Principle – the logic behind it

3.1 The system as a chain

An optical transmission system consists of an **electro-optical converter** (LED/laser), a **transmission path** (air, glass fibre) and an **opto-electrical converter** (photodiode). The chain formula $I_A = \eta_{LED} \cdot \kappa \cdot \eta_{PD} \cdot I_E$ (in dB the terms add) explains every measurement: η are conversion efficiencies, κ the path transmission.

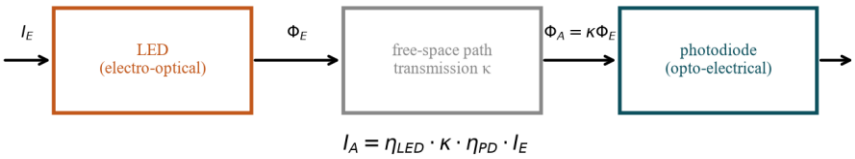


Fig. 1: Block diagram of the transmission system.

3.2 LED and photodiode

Both are pn junctions: $I = I_S(e^{U/U_T} - 1)$. In the **LED** injected carriers recombine and emit photons; the forward voltage is ≈ W_G/e. In the **photodiode** the field of the space-charge zone separates optically generated pairs: the photocurrent $I_{ph} = S \cdot P$ is proportional to the light power and shifts the curve down by I_{ph} ($I_{SC} = -I_{ph}$; in the 4th quadrant solar-cell operation). $I_{ph}(P)$ is linear in reverse bias/short circuit and without saturation; a large load resistance (here 20 kΩ) reduces linearity and bandwidth.

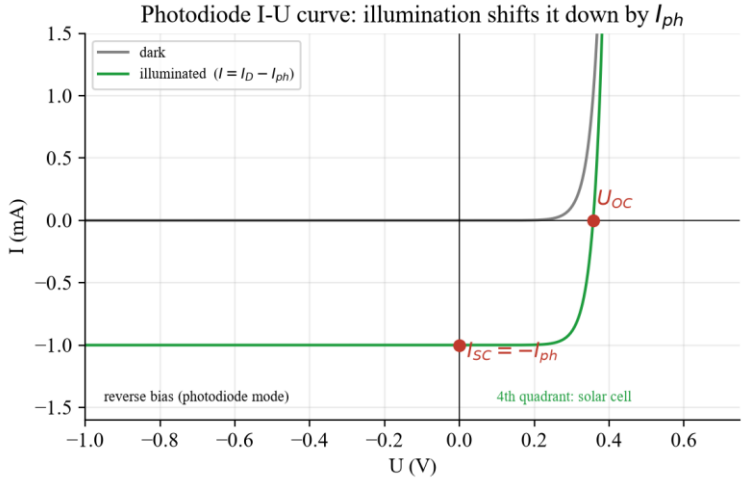


Fig. 2: Photodiode curve dark and illuminated (model).

3.3 Distance and radiation

Far from the LED the irradiance falls as 1/r². In the experiment r is only 0 … 2.5 cm (the size of the 5 mm components): $I_{PD} \approx I_0/(1 + r/r_0)^2$; plotting $I_{PD}^{-1/2}$ against r gives a straight line with the intercept $-r_0$. An LED with a **Lambert pattern** (60°, diffuse housing) radiates broadly, a **lens LED** (12°) concentrates the same power into a small solid angle (≈ 10–25 times the on-axis intensity).

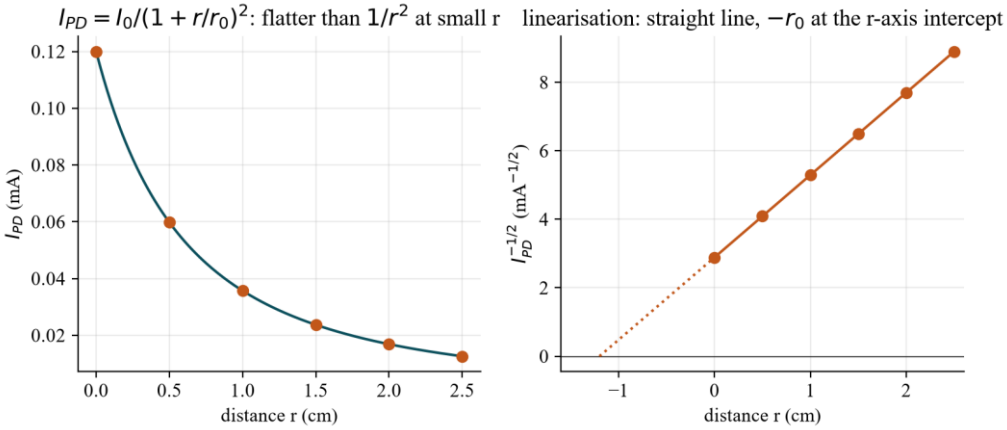


Fig. 3: Model for r = 0 … 2.5 cm and linearisation.

3.4 Dynamics: the RC low-pass

Capacitances and the series resistor form an RC low-pass: step response $1 - e^{-t/\tau}$, rise time (10–90 %) $\mathbf{t_r = 2.2\ \tau}$, cut-off frequency $\mathbf{f_{gr} = 1/(2\pi\tau) = 0.35/t_r}$. In the frequency domain the amplitude is −3 dB at f_{gr}, the phase is $\varphi = \arctan(f/f_{gr})$ and the delay $t_0 = \varphi/(360° \cdot f)$ – constant ≈ RC at low frequency, ∝ 1/f at high frequency.

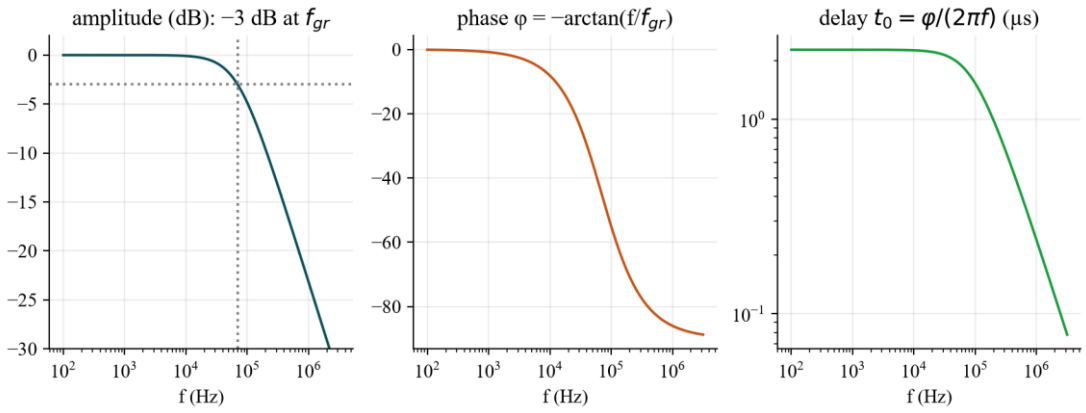


Fig. 4: Low-pass: amplitude, phase, delay.

4. Procedure step by step

Task	What to do?	Evaluation
a) I-U	U_{source} = −1 … 7 V (fine at 1.2–1.8 V); read U_{LED} and I_{LED}	$I(U_{LED})$, forward voltage U_F
b) PD	r = 0 … 2.5 cm (0.5 cm steps), U_{supply} = 7 V, R = 20 kΩ; measure U_{PD}	I = U/R, P = U²/R, dBm
c) η	$P_{LED} = U_{supply} \cdot I_{LED}$, P_{PD} at r_{min}	$\eta = P_{PD}/P_{LED}$ (≈ 10^{-4})
d) t_r, t_f	Square wave 20 kHz, offset 5 V, amplitude 5 V; CH1 before, CH2 after the system	10–90 %, $f_{gr} = 0.35/t_r$
e) t_0, φ	Sine 1 … 80 kHz; time shift between CH1 and CH2	$\varphi = 360° \cdot f \cdot t_0$
f) f_{3dB}	Raise the sine from 250 Hz until −3 dB	Bode diagram, f_{3dB}
g) Audio	LED + PD, ≥ 1 m, align, bias/offset	Describe analog vs. digital

Optional: LTspice demo of the RC low-pass (rise time): Freistrahl_RC_Tiefpass.asc (or *04_LTspice/LTspice_starten.bat*).

5. What to watch in the lab

- Bring: calculator, pencil, **30 cm ruler, linearly divided graph paper**.
- Mind the current limit of the LED (high voltages only with the series resistor).
- Ambient light (100 Hz from lamps) disturbs the photodiode – shield it.
- At small distances the photodiode may saturate (large U_{PD}) – check linearity.
- Measure the rise time between 10 % and 90 % of the final value; make sure the signal reaches it within the half period.
- The written elaboration is done after the lab; all measurements must be finished on site.

6. Why do we need this later?

- **Experiment 4:** LED characteristic, radiation pattern and efficiency are deepened there.
- **Experiment 2:** the photodiode in the 4th quadrant is the solar cell.
- **Experiments 5/6:** laser diodes as transmitters; bandwidth and linearity are the same questions.
- Practice: remote controls, light barriers, fibre and free-space communication (Li-Fi).

7. Typical questions – short answers

Question	Short answer
Three elements of the system?	Electro-optical converter, path, opto-electrical converter.
Chain formula?	$I_A = \eta_{LED} \cdot \kappa \cdot \eta_{PD} \cdot I_E$.
Illuminated photodiode?	Curve shifted down by I_{ph}: $I_{SC} = -I_{ph}$, $U_{OC} = U_T \ln(I_{ph}/I_S + 1)$.
When is I ∝ P?	Photodiode in reverse/short circuit without saturation; LED at moderate currents.
Distance law?	Far field $1/r^2$; here near: $I_0/(1 + r/r_0)^2$.
Beam angle?	Angle where the intensity falls to ½: 60° (Lambert) / 12° (lens).
Why $\eta \approx 10^{-4}$?	LED conversion × geometric coupling × PD conversion; small power in the 20 kΩ resistor.
t_r and f_{gr}?	$t_r = 2.2\ \tau$; $f_{gr} = 0.35/t_r$.
Phase and delay?	$\varphi = \arctan(f/f_{gr}) = 360° \cdot f \cdot t_0$.
−3 dB frequency?	Amplitude falls to 0.708 of the value at 250 Hz.
Why offset 5 V?	The LED conducts only forward: 0…10 V keeps the current positive.
Analog vs. digital?	Analog: continuous intensity modulation, noise adds; digital: bits with threshold decision, robust, needs bandwidth.

Date: **20 Jan 2027**, pre-test (Antestat) 14:00, building LT, room LT 227 (group 9).